

CANDIDATE NAME _____

INDEX NUMBER _____

CG _____



SERANGOON JUNIOR COLLEGE
JC2 Preliminary Examination 2015

H2 BIOLOGY
9648

Paper 2

2 Hours

Additional materials:
Writing papers

Date / Day: 21 AUG 2015 / Friday

INSTRUCTIONS TO CANDIDATES

Write your name, CG and index number in the spaces at the top of this page and on all separate writing papers used.

Write in dark blue or black pen.

You may use a soft pencil for any diagrams, graphs or rough working.

Section A

Answer **all** questions.

Write your answers in spaces provided on the question paper.

Section B

Answer **only one** question out of two.

Write your answers on the separate answer paper provided.

INFORMATION FOR CANDIDATES

The intended number of marks is given in brackets [] at the end of each question or part question.

FOR EXAMINER'S USE	
Section A	
1	/9
2	/10
3	/9
4	/12
5	/10
6	/8
7	/12
8	/10
Total	/80
Section B	
9 OR 10	/20
TOTAL	/100

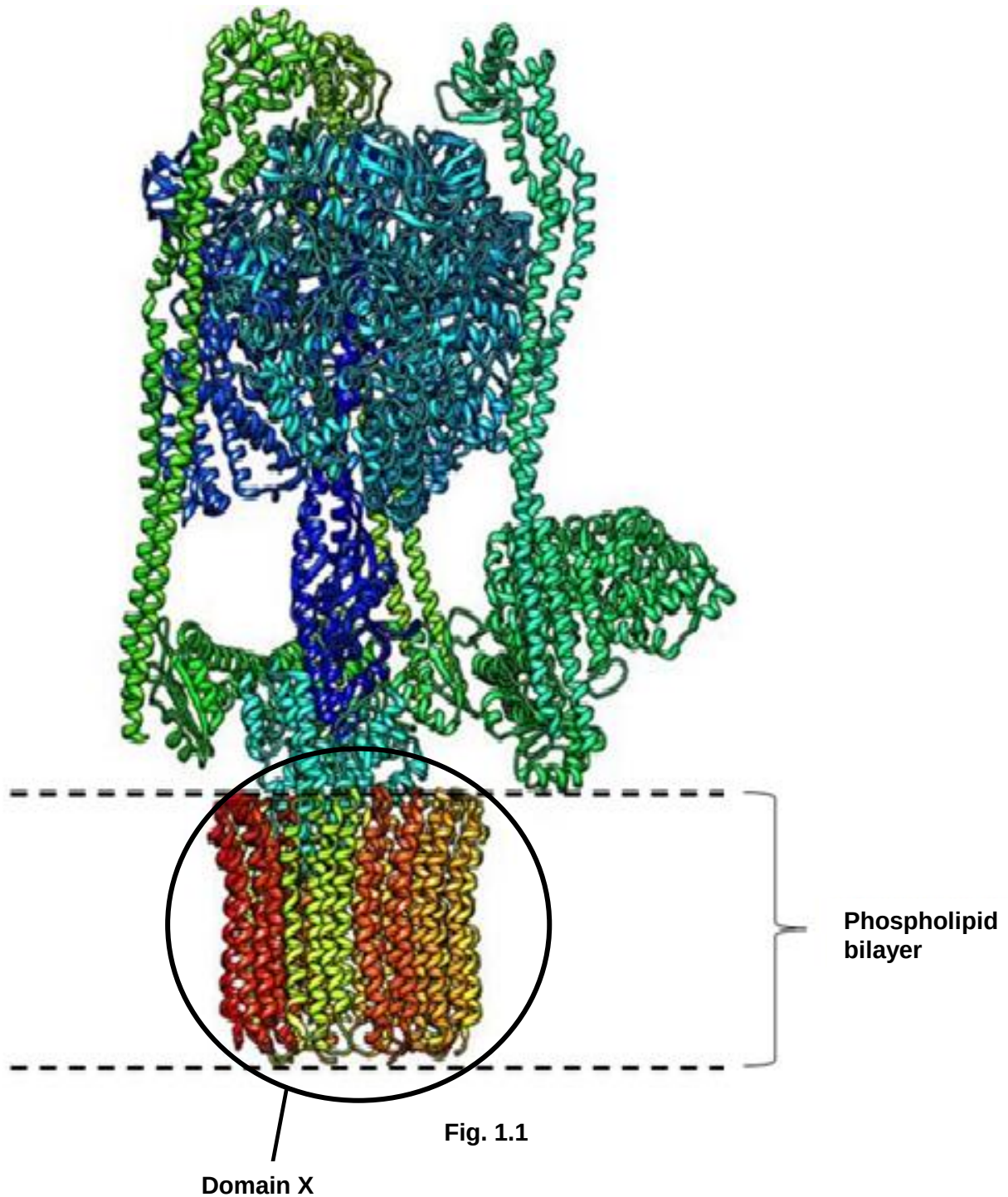
This question paper consists of 23 printed pages and 1 blank page

SECTION A

Answer **all** questions.

Question 1

Transport proteins are embedded within the membranes of various organelles. **Fig. 1.1** shows the structure of one such transport protein. The protein is known to contain multiple polypeptide chains.



(a) With reference to **Fig. 1.1**, describe how the structure of Domain X is maintained. [2]

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A researcher intends to study the transport of substances carried out by the transport protein. The researcher begins by creating artificial vesicles containing only the transport protein embedded within the membrane. The artificial vesicles were then placed in 5 different solutions, each containing a different substance with a concentration of 10 arbitrary units.

The concentrations of the substances in the artificial vesicles were measured before the experiment and after 15 minutes of the experiment. The results obtained by the researcher are shown in **Table 1.2**.

Solution	Concentration of substances in the artificial vesicles before experiment	Concentration of substances in the artificial vesicles after 15 minutes of the experiment
1	0 AU of Na ⁺ ions	0 AUs of Na ⁺ ions
2	0 AUs of Cl ⁻ ions	0 AUs of Cl ⁻ ions
3	1 AUs of H ⁺ ions	20 AUs of H ⁺ ions
4	0 AU of protein insulin	0 AU of protein insulin
5	0 AU of amino acid glutamate	0 AUs of acidic amino acid glutamate

Table 1.2

(b) With reference to **Table 1.2**, explain the role of the transport protein in transporting substances across membranes. [4]

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Another Protein A is a receptor protein. However, unlike the transport protein shown in **Fig. 1.1**, it is made up of only a single polypeptide and is embedded within the cell surface membrane.

- (c) Outline the route taken by Protein A after it is synthesized by the ribosome on the rough endoplasmic reticulum. [3]

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[Total: 9]

Question 2

Fig. 2.1 shows cultured fibroblasts of the Indian barking deer, *Muntiacus muntjac*, undergoing different stages of the mitotic cell cycle, **J** to **Q**.

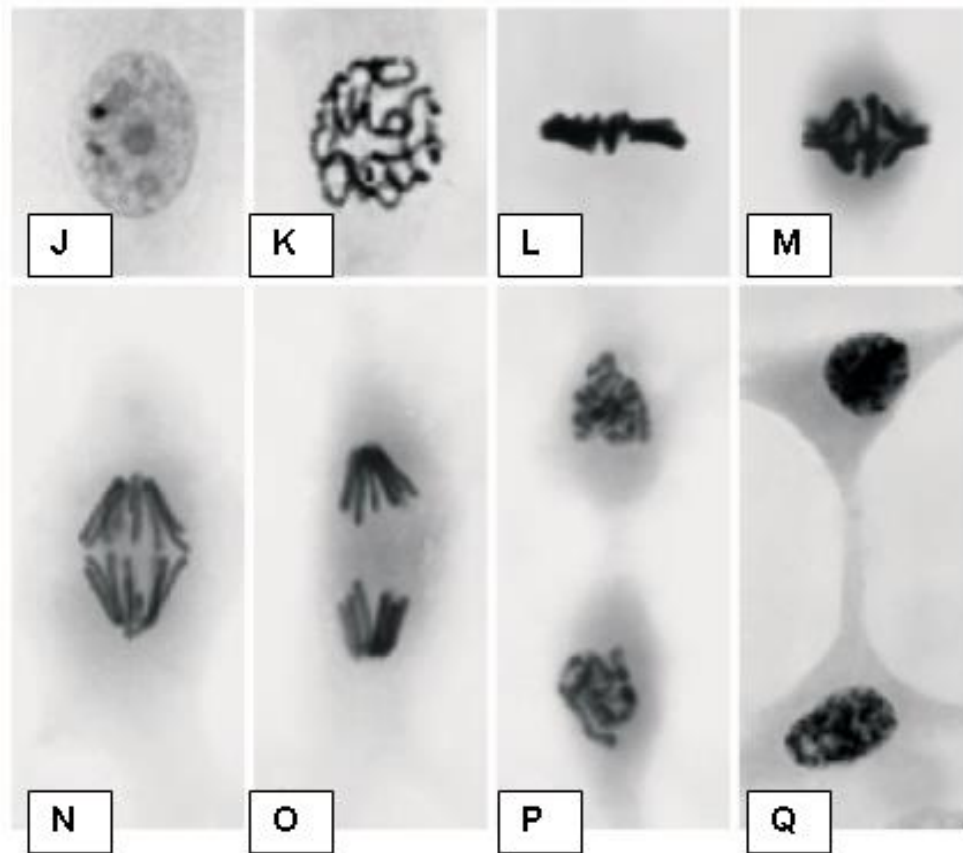


Fig. 2.1

(a) With reference to **Fig. 2.1**, identify stages **K**, **L**, **N** and **Q**. [2]

K: **L:**

N: **Q:**

(b) Describe the events that occur in stage **P**. [2]

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(c) Outline the roles of centromeres in mitosis. [3]

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(d) Fig. 2.2 shows changes in DNA content at different stages of the animal life cycle.

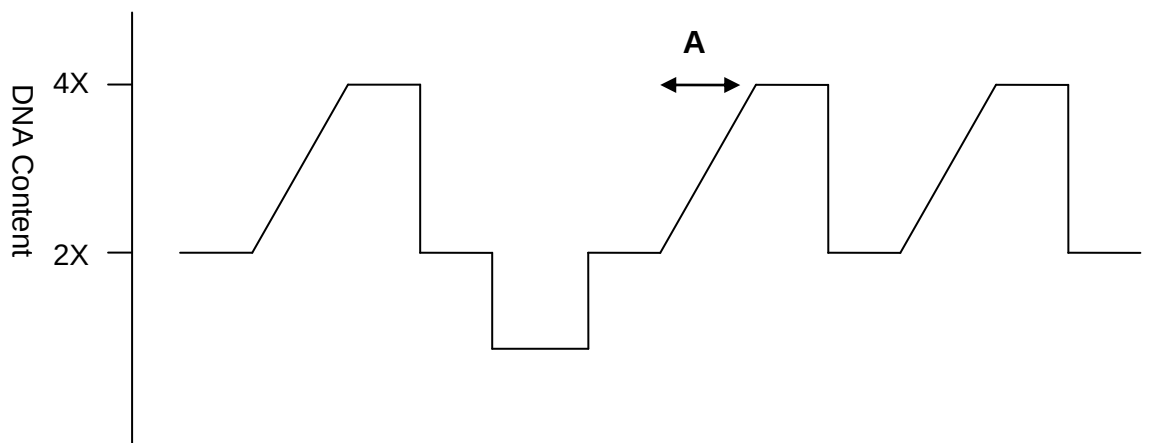


Fig. 2.2

With reference to Fig. 2.2,

(i) identify and describe Stage A. [2]

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(ii) state the significance of Stage A. [1]

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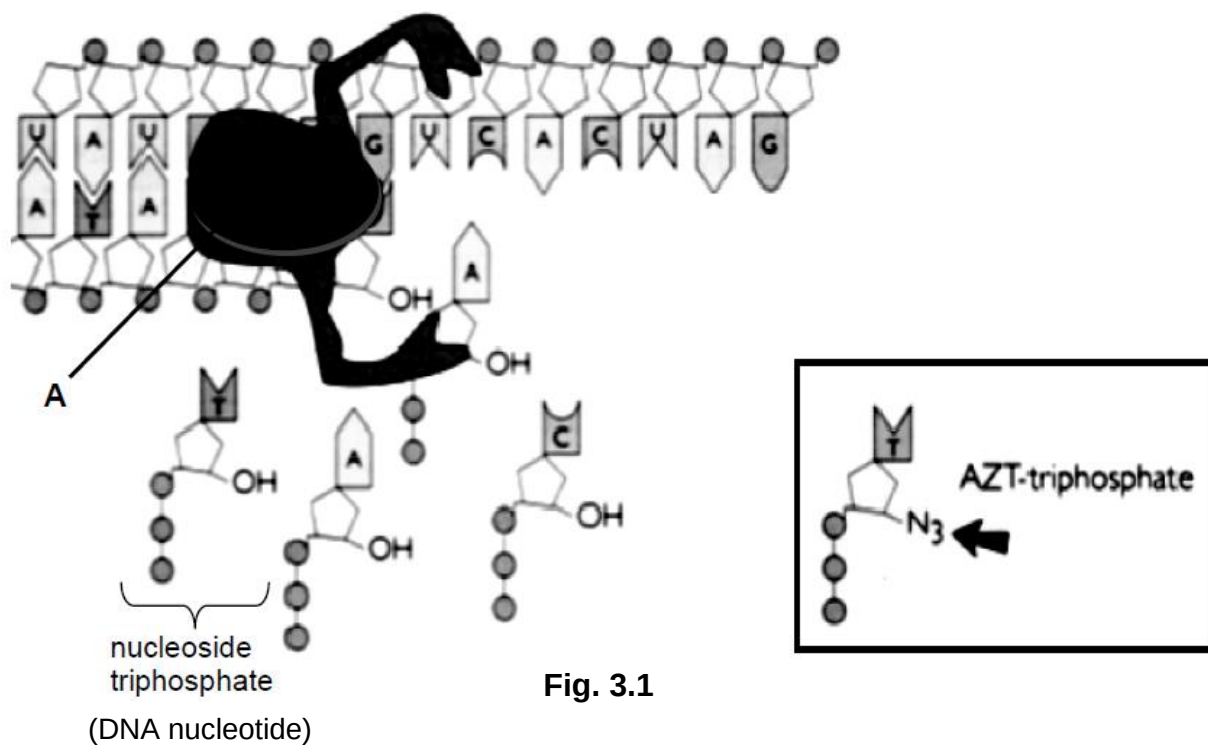
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[Total: 10]

Question 3

- (a) The Human Immunodeficiency Virus (HIV) causes Acquired Immune Deficiency Syndrome (AIDS).

Fig. 3.1 shows the molecular process in an infected host cell that has the anti-viral drug AZT-triphosphate introduced.



- (i) Name enzyme **A** and describe the process it catalyses. [2]

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- (ii) Explain the action of AZT-triphosphate as an anti-viral drug. [3]

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- (iii)** With reference to the structure of A, suggest one way in which AZT-triphosphate loses its effectiveness as an anti-viral drug. [2]

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- (b)** Describe how the reproductive cycles of lambda phage differs from HIV. [2]

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[Total: 9]

Question 4

- (a) **Fig. 4.1** illustrates the structure of a telomere. The looped segment of the telomere shown is the T loop.

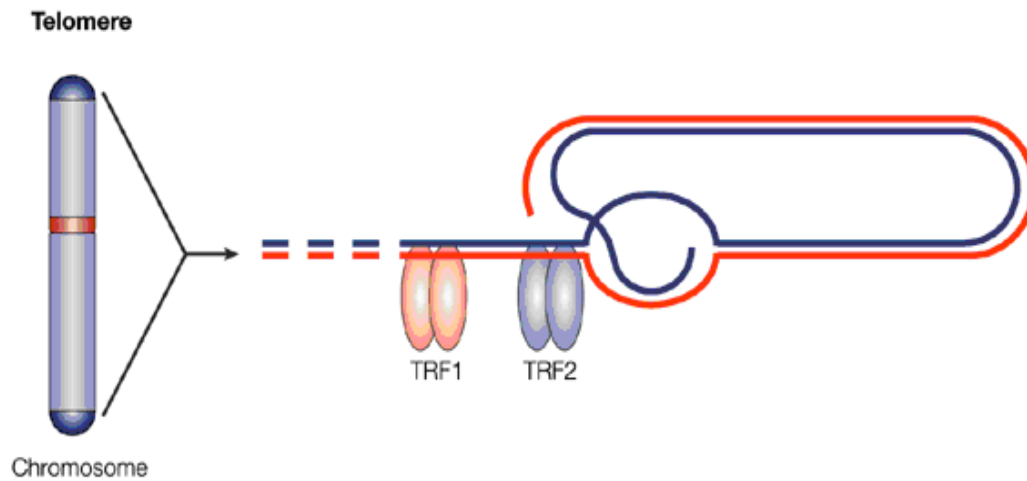


Fig. 4.1

- (i) Label on the diagram, the 3' and 5' ends of the DNA. [1]
- (ii) Explain the phenomenon that enabled you to distinguish the 3' and 5' ends of the DNA in part (a) (i). [3]

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- (b) Describe the role of telomeres. [2]

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Fig. 4.2 shows the growth pattern of bacteria cells in a medium containing a mixture of glucose and lactose. Observations of the concentrations of the two sugars in the medium show that glucose falls to very low levels after the start of the experiment, but lactose remains relatively high and only starts to decrease near the end of the experiment (**not shown in Fig. 4.2**).

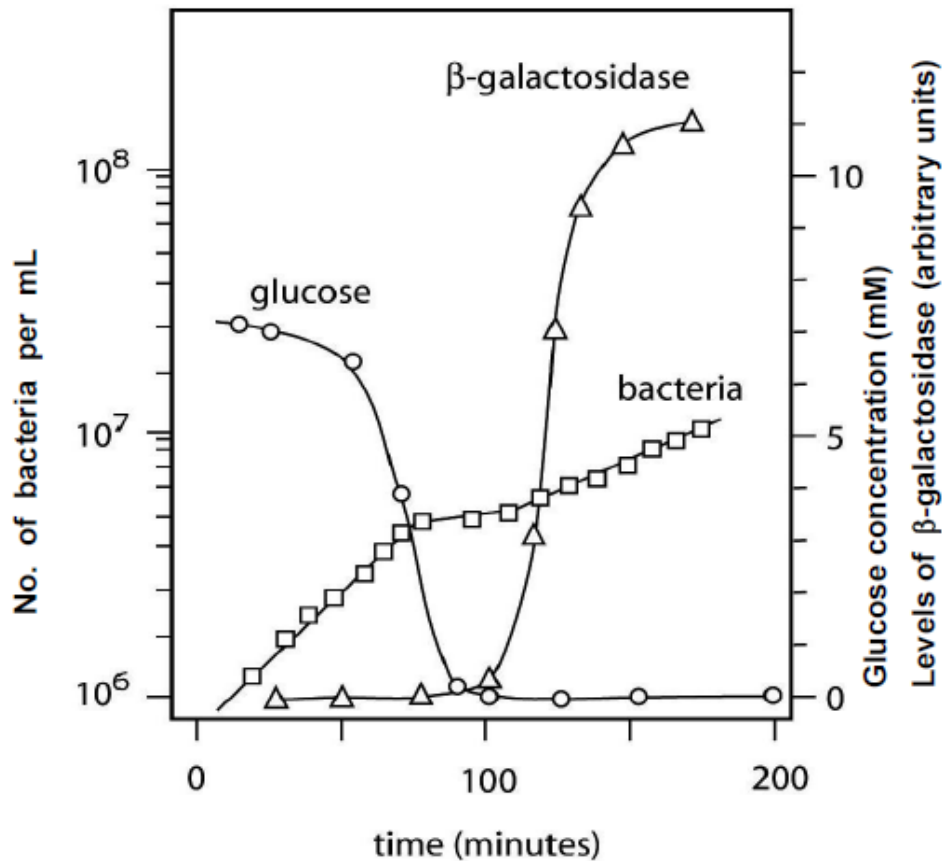


Fig. 4.2

From the experiment, it was observed that the growth rate pattern of bacteria cells showed a rapid increase in number, constant numbers in the middle of the experiment followed by a slower increase towards the end.

- (c) Suggest why the number of bacteria cells remained constant during the middle of the experiment. [1]

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- (d) With reference to **Fig. 4.2**, explain why β -galactosidase levels only increased after 100 minutes. [3]

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In prokaryotes, genes are often organized into operons.

- (e) Using named examples, discuss the differences between inducible and repressible operons. [2]

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[Total: 12]

Question 5

A group of genetics students was investigating the inheritance of body colour and wing shape in a new species of fly. Pure-breeding flies with wild-type bodies and vestigial wings were crossed with pure-breeding flies with black bodies and wild-type wings.

All the F_1 offspring had wild-type bodies and wild-type wings.

Flies from the F_1 were crossed with black bodied, vestigial-winged males. 2876 offspring flies were produced as shown below:

wild-type body, vestigial wings	1277
wild-type body, wild-type wings	143
black body, vestigial wings	152
black body, wild type wings	1304

- (a) Based on the information and results given above, describe the inheritance of body colour and wing shape in this species of fly. [2]

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- (b)** Draw a genetic diagram to explain the observed results of the cross between F_1 flies and black bodied, vestigial-winged flies.

Use the following symbols to represent the different alleles involved:

A/a for gene locus controlling body colour,

B/b for gene locus controlling wing shape.

[4]

- (c) The genetics students then investigated the inheritance of body length of the fly. Pure-breeding flies with adult body length of more than 3.8 mm were crossed with pure-breeding flies with adult body length of less than 2.4 mm.

It was observed that the F_1 offspring showed a range of adult body lengths with most flies having a body length of 3.0 mm. When the F_1 offspring were crossed with each other, the F_2 offspring were observed to have a wider range of adult body length than the F_1 offspring.

- (i) State the term used to describe the range of body lengths in the F_1 and F_2 offspring. [1]

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- (ii) Explain why a wide range of body lengths was observed in the F_2 offspring compared to the F_1 offspring. [3]

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[Total: 10]

Question 6

- (a) **Fig. 6.1** shows the cell signalling pathway by hormone Epidermal Growth Factor (EGF).

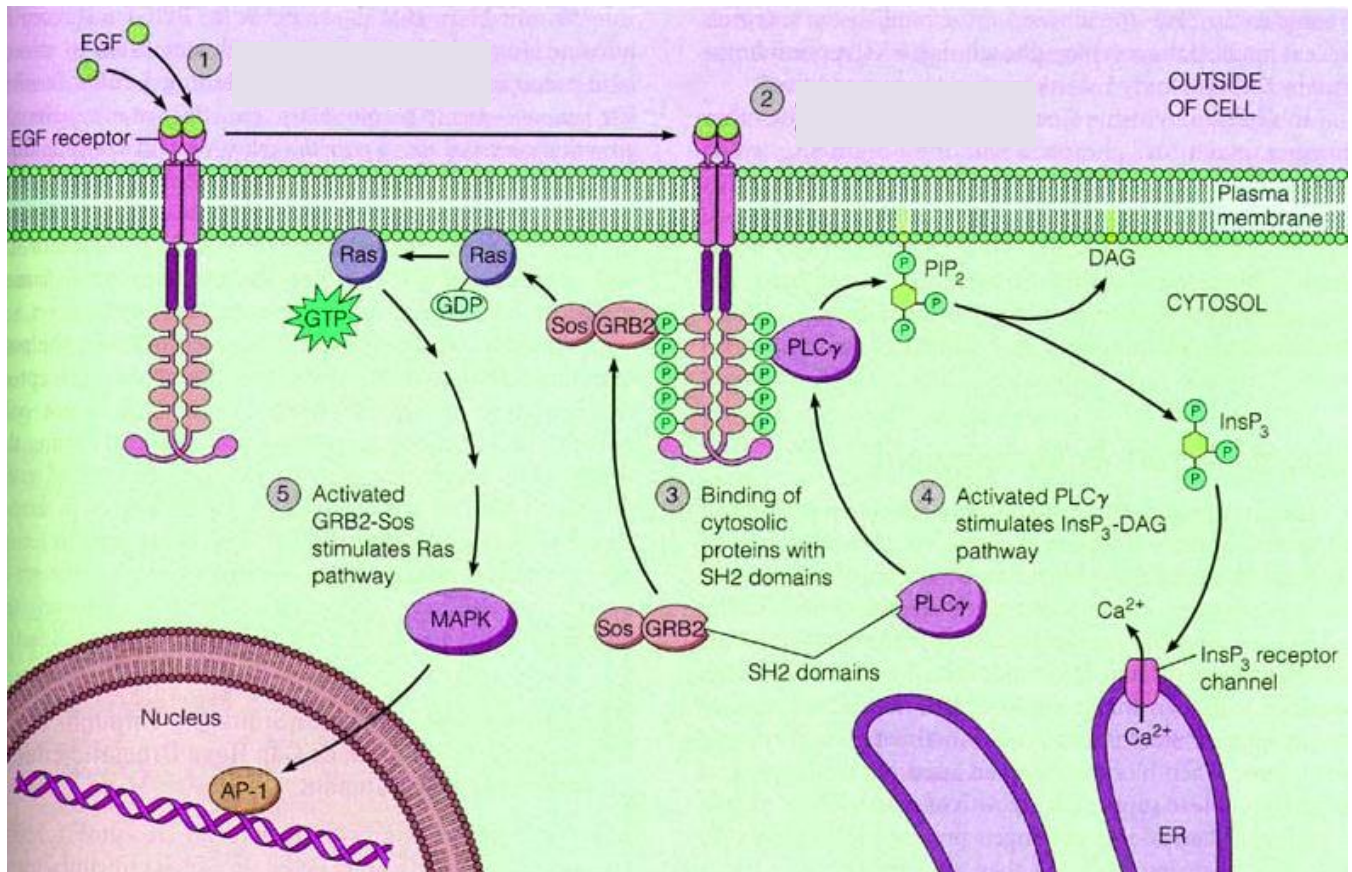


Fig. 6.1

- (i) With reference to **Fig. 6.1**, describe the events occurring in steps 1 and 2. [3]

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- (ii) When Ras protein is activated, it could result in large numbers of activated MAP Kinase (MAPK). Suggest what could happen during stage 5 that would lead to such an outcome. [2]

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- (iii) The receptor for EGF can be activated by a deletion that removes part of its extracellular domain. These mutant receptors are able to form active dimers even in the absence of EGF, and thus they produce a stimulatory signal inappropriately. Besides mutation to the gene that codes for the extracellular domains of the receptors, 30% of human tumors have a Ras mutation.

With reference to **Fig. 6.1**, suggest and explain how a missense mutation in the Ras gene may bring about cancer. [3]

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[Total: 8]

Question 7

- (a) **Fig. 7.1** shows the changes in membrane potential at one point on an axon of a nerve cell.

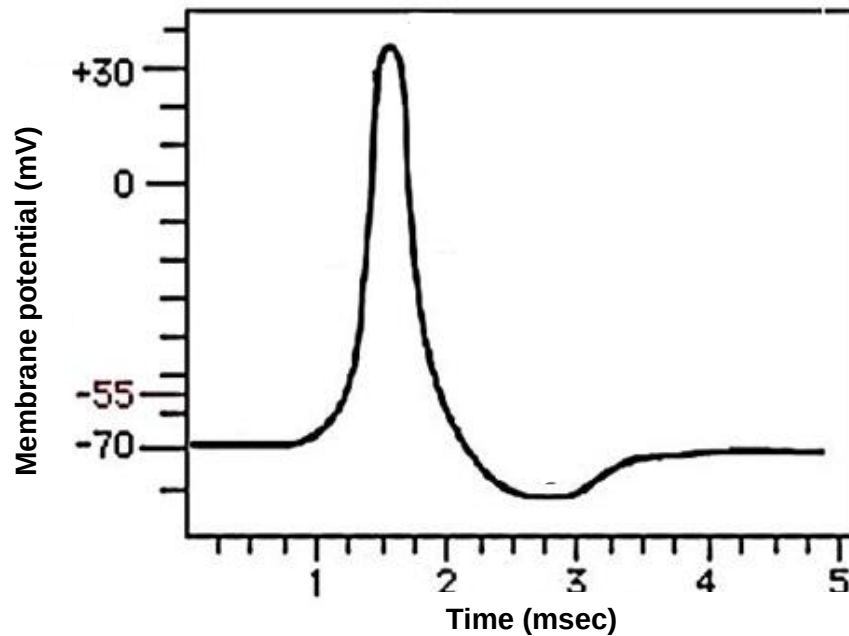


Fig. 7.1

With reference to **Fig. 7.1**, explain the shape of the graph

- (i) between 1.0 msec and 1.5 msec [2]

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- (ii) between 2.5 msec and 3.0 msec [2]

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- (b) Suggest why transmission of impulses along a myelinated axon uses less energy than an unmyelinated axon of the same diameter. [2]

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- (c) A new painkiller that is much more effective than morphine became available in the United Kingdom in 2006. It was extracted from the venom of the Magician's Cone Snail, *Conus magus*.

C. magus is a native of coral reefs in the Pacific Ocean. It injects the venom into fish, which are paralysed and swallowed whole. Scientists have discovered that the venom blocks calcium channels at synapses between neurones.

Explain how the blocking of calcium channels can result in both reduction of pain in humans and paralysis in fish. [3]

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- (d) Diabetes mellitus is often due to the destruction of the β -cells in the islets of Langerhans in the pancreas. **Fig. 7.2** shows the changes in blood glucose concentration over a day in which an untreated diabetic person and a non-diabetic person take the same meals and exercise. Blood glucose was measured in both individuals every hour. At meal times, the measurements were made just before the meal.

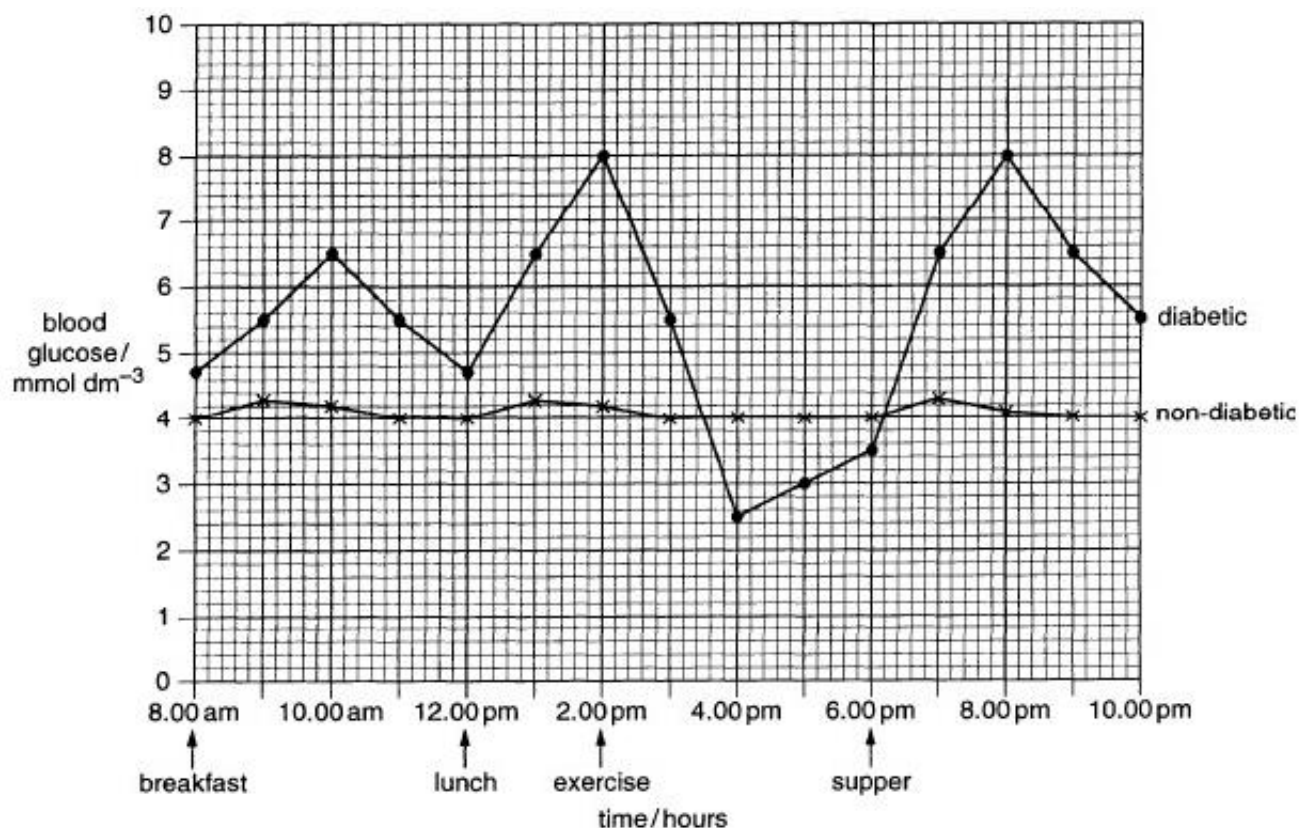


Fig. 7.2

Explain why, following exercise at 2.00 pm,

- (i) the blood glucose in the diabetic person dropped; [1]

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- (ii) the blood glucose in the non-diabetic person stayed the same. [2]

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[Total: 12]

Question 8

A population of a diploid species inhabiting remote island X was tracked for many generations. At the start of the tracking, it was found that a particular locus of one chromosome can be occupied by one of two sequences – Sequence A or Sequence B. These two sequences were observed to be non-coding.

The population size of 100 of the species has been roughly stable throughout the generations in island X. The number of individuals possessing Sequence A and Sequence B at the start of the tracking is shown in **Table 8.1**.

Sequence possessed	Number of individuals
Sequence A only	25
Sequences A and B	50
Sequence B only	25

Table 8.1

- (a) In the space below, calculate the frequency of Sequence A in the population at the start of the tracking. [1]

Fig. 8.2 shows the change in the frequency of Sequence A over many generations in the population on island X.

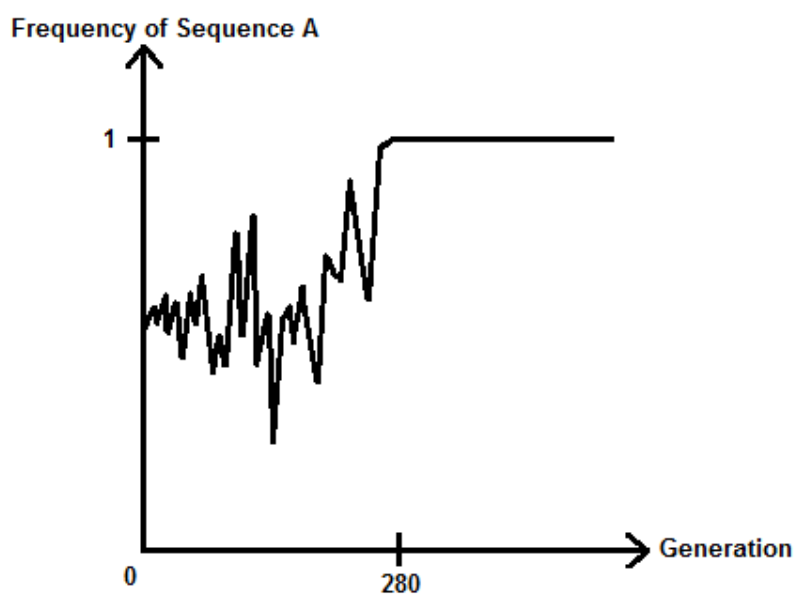


Fig. 8.2

- (b) With reference to **Fig. 8.2**, describe the change in the frequency of Sequence A. [2]

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- (c) Provide an explanation for the change in the frequency of Sequence A over the generations. [3]

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The same species was also found to inhabit another remote island Y with a population size of 2000. The same two sequences of A and B were also found to occupy the same locus in this population. **Fig. 8.3** shows the change in the frequency of Sequence A over many generations in the population on island Y.

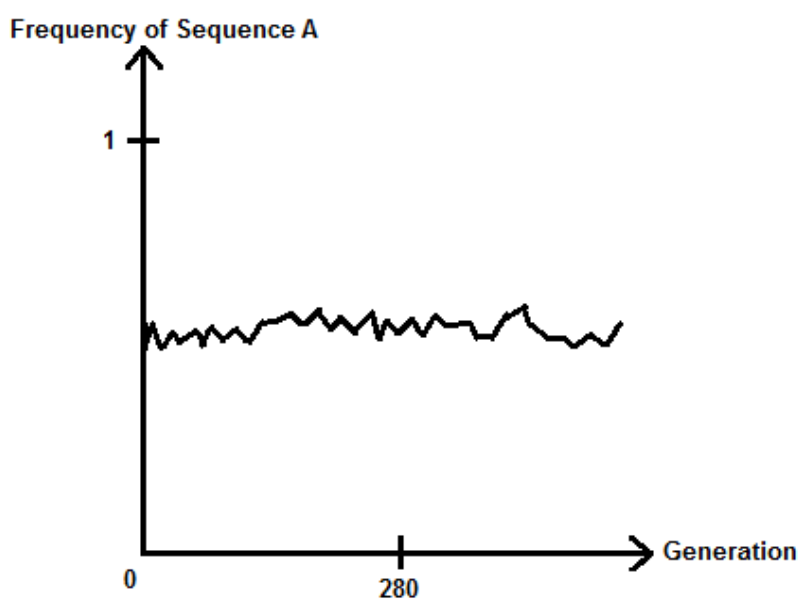


Fig 8.3

- (d)** Suggest and explain the differences in the frequency of the sequence A over many generations in the two populations on island X and Y. [2]

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A related species Y was discovered. To determine if species X and Y are related, an investigator studied DNA sequences which codes for an important protein that performs the same function in both X and Y.

Sequence D (from species X) and Sequence E (from species Y) were identical except at two points.

- (e)** Explain how Sequences D and E provide evidence to support the theory of evolution. [2]

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[Total: 10]

Section B

Answer **one** question

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labeled diagrams, where appropriate.

Your answers must be in continuous prose where appropriate.

Your answers must be set out in sections (a), (b), etc as indicated in the question.

Question 9

- (a) Explain why and how eukaryotic pre-mRNA is processed. [6]
- (b) Compare the two processes that have to take place for protein synthesis. [9]
- (c) Explain how genetic variation arises in sexually reproducing organisms. [5]

Question 10

- (a) Outline the main features of oxidative phosphorylation and contrast this with photophosphorylation. [9]
- (b) Compare the Krebs cycle and the Calvin cycle. [6]
- (c) Outline the process of fermentation in yeasts and explain its significance. [5]

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